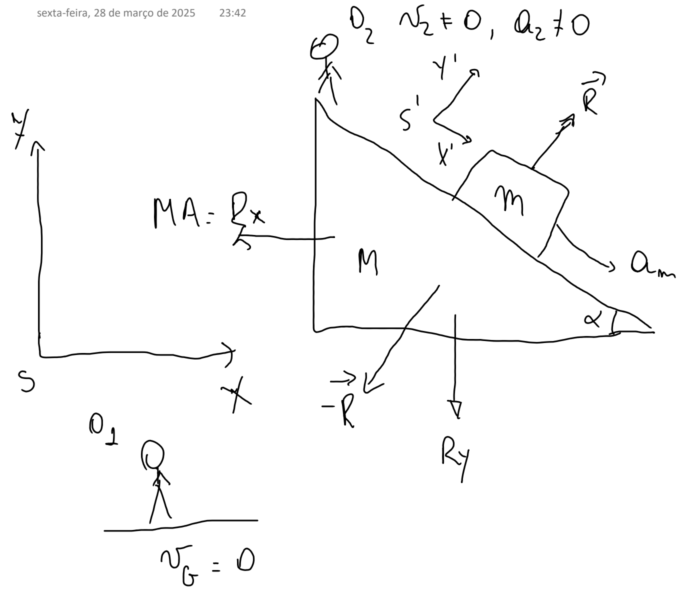
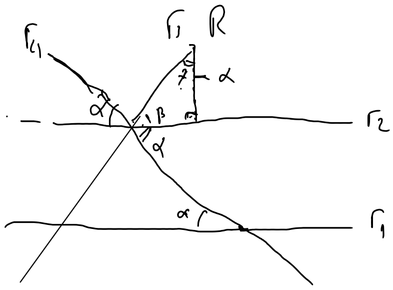


I) NO FRICTION FORCES BETWEEN THE BLOCK AND THE WEDGE

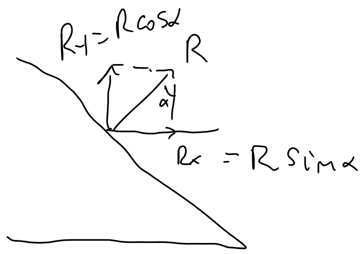
II) THE WEDGE MOVES WITHOUT FRICTION

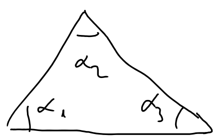
Obs::

i) a_m is the relative acceleration between the block and the wedge



$r_2 // r_1$, $\alpha + \beta = \frac{\pi}{2} \Rightarrow \beta = \frac{\pi}{2} - \alpha$



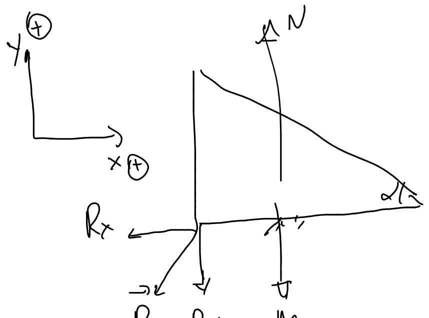


$\alpha_1 + \alpha_2 + \alpha_3 = \pi$

$\beta + x + \frac{\pi}{2} = \pi$

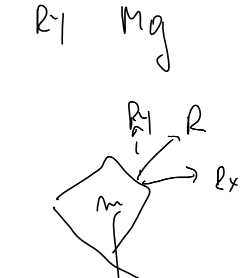
$\frac{\pi}{2} - \alpha + x + \frac{\pi}{2} = \pi$

$\Rightarrow x = \alpha$



$$\begin{cases} -R_x = MA_x = -R\sin\alpha \\ N - mg - R_y = MA_y = 0 \end{cases}$$

$$\boxed{\begin{aligned} R\sin\alpha &= -MA_x \\ N &= mg + R\cos\alpha \end{aligned}}$$



$$\begin{cases} R_x = ma_x \\ R_y - mg = ma_y \end{cases}$$

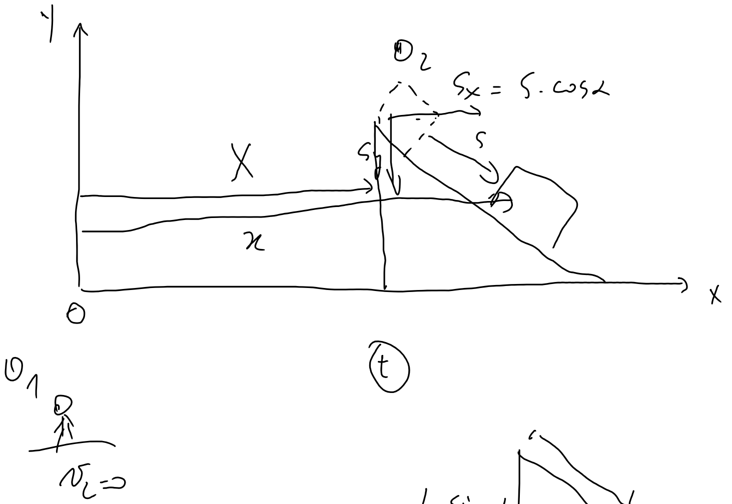
$\boxed{a_y < 0,\ a_y = 0,\ a_y > 0}$?

$$(1)\quad R\sin\alpha = -MA_x$$

$$(2)\quad R\sin\alpha = m a_x$$

$$(3)\quad R\cos\alpha - mg = m a_y$$

R, A_x, a_x, a_y
4? 3 eq.



$$x = X + s\cdot\cos\alpha$$

$$y = L\sin\alpha - s\cdot\sin\alpha$$

$x = x(t)$, $X = X(t)$, $s = s(t)$
$y = y(t)$

$$\begin{cases} \ddot{x} = \ddot{X} + \ddot{s}\cdot\cos\alpha \\ \ddot{y} = -\ddot{s}\cdot\sin\alpha \end{cases}$$

$$\ddot{x} = a_x,\ \ddot{y} = a_y,\ \ddot{X} = A_x,\ \ddot{s} = a_m = a_{b/w}$$

$$a_x = A_x + \ddot{s}\cdot\cos\alpha \Rightarrow \ddot{s} = (a_x - A_x)\cdot\frac{1}{\cos\alpha}$$

$$a_y = -\ddot{s}\cdot\sin\alpha \Rightarrow \ddot{s} = -\frac{a_y}{\sin\alpha}$$

$$\Rightarrow \boxed{a_y = (A_x - a_x)\,tg\,\alpha}$$

$$\left.\begin{array}{l}(1)\quad R\sin = -MA_x \\ (2)\quad R\sin\alpha = m a_x\end{array}\right\} \longrightarrow \boxed{A_x = -\frac{m}{M}a_x}$$

$$(3)\quad R\cos\alpha = m(a_y + g) \longrightarrow a_y = \frac{R\cos\alpha}{m} - g$$

$$(4)\quad a_y = (A_x - a_x)\,tg\,\alpha$$

$$\begin{cases} a_y = \left(-\frac{m}{M}a_x - a_x\right)tg\,\alpha = -\left(\frac{m+M}{M}\right)a_x\,tg\,\alpha \\ a_y = \frac{R\cos\alpha}{m} - g = \frac{m a_x}{m\sin\alpha}\cos\alpha - g = a_x\,cotg\,\alpha - g \end{cases}$$

$$a_y = a_x\,cotg\,\alpha - g = -\left(\frac{m+M}{M}\right)a_x\,tg\,\alpha$$

$$a_y = a_x \operatorname{cotg}\alpha - g = -\left(\frac{m+M}{M}\right) a_x \operatorname{tg}\alpha$$

$$M a_x \cos^2\alpha - M g \cos\alpha \sin\alpha = -(m+M)\, a_x \sin^2\alpha$$

$$(\underbrace{M\cos^2\alpha + M\sin^2\alpha} + m\sin^2\alpha)\, a_x = Mg\cos\alpha\sin\alpha$$

$$\boxed{a_x = \frac{Mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha}}$$

$$\boxed{A_x = -\frac{m}{M} a_x = -\frac{mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha}}$$

$$a_y = (A_x - a_x)\operatorname{tg}\alpha = -\frac{g\sin\alpha\cos\alpha}{M + m\sin^2\alpha}(m+M)\operatorname{tg}\alpha$$

$$\boxed{a_y = -\frac{(m+M)}{M + m\sin^2\alpha}\, g\sin^2\alpha}$$

a_x, a_y, A_x

$$\ddot{s} = a_m = -\frac{a_y}{\sin\alpha} = \frac{(m+M)}{M + m\sin^2\alpha}\, g\sin\alpha$$